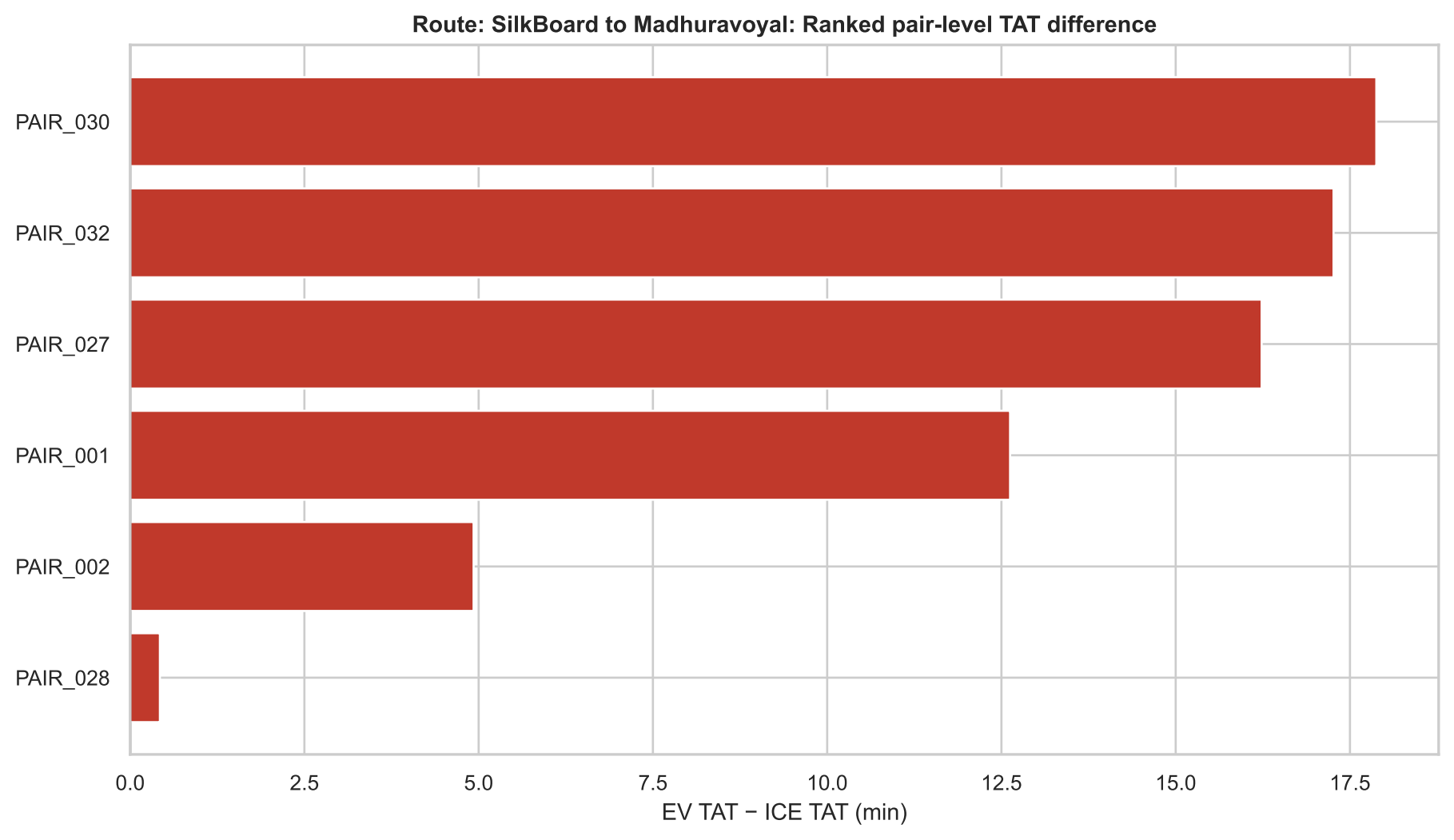
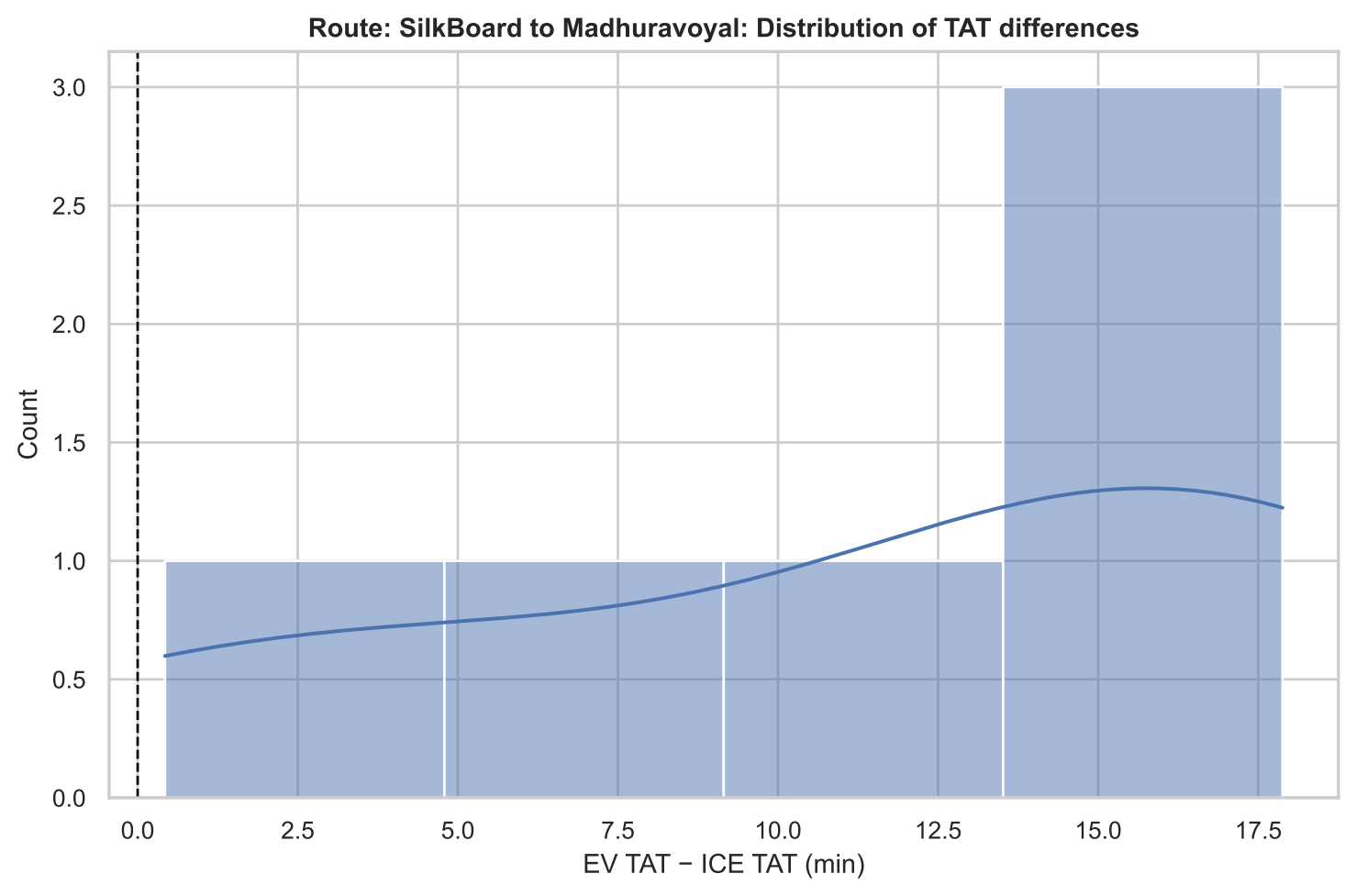
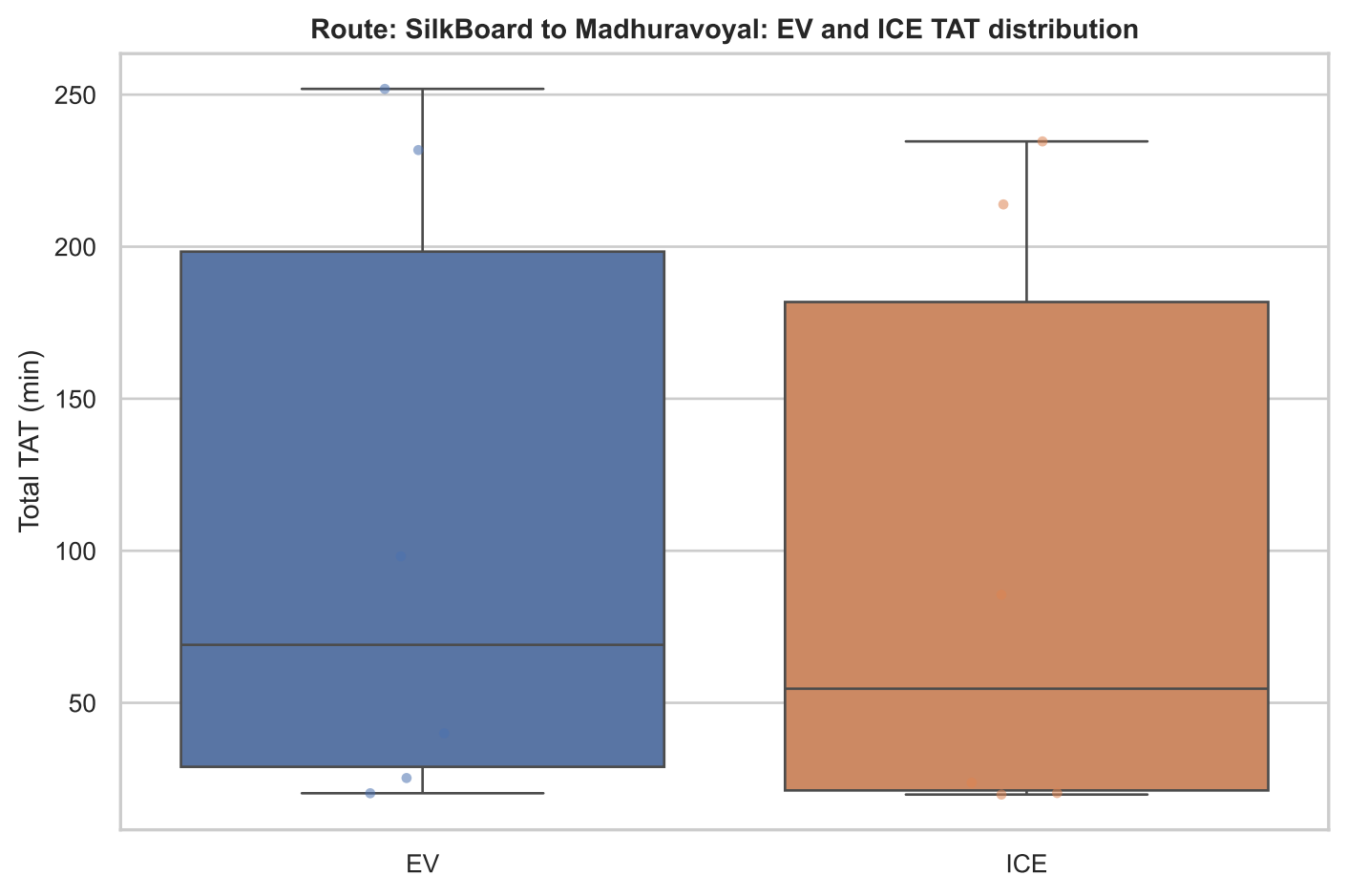
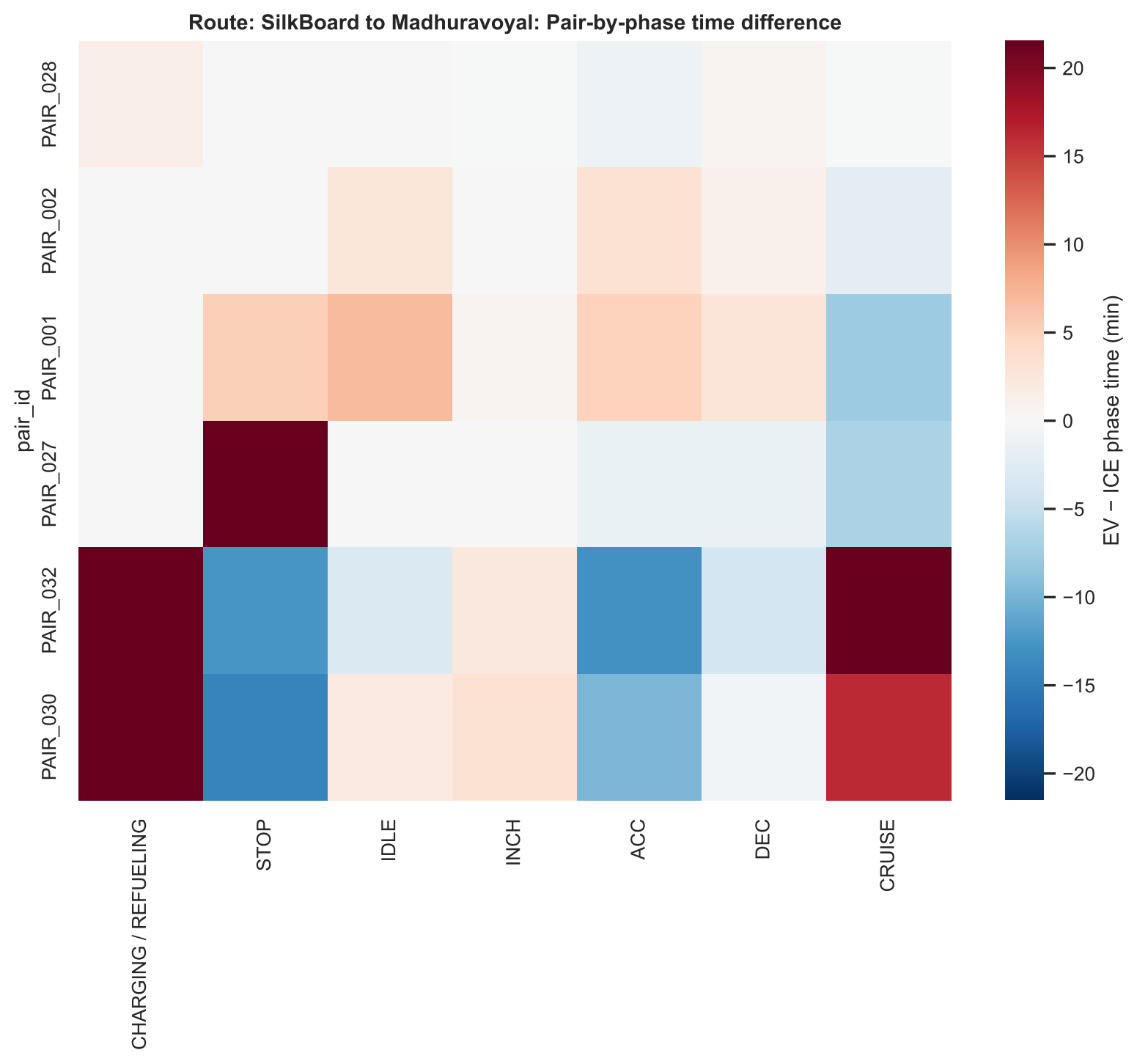
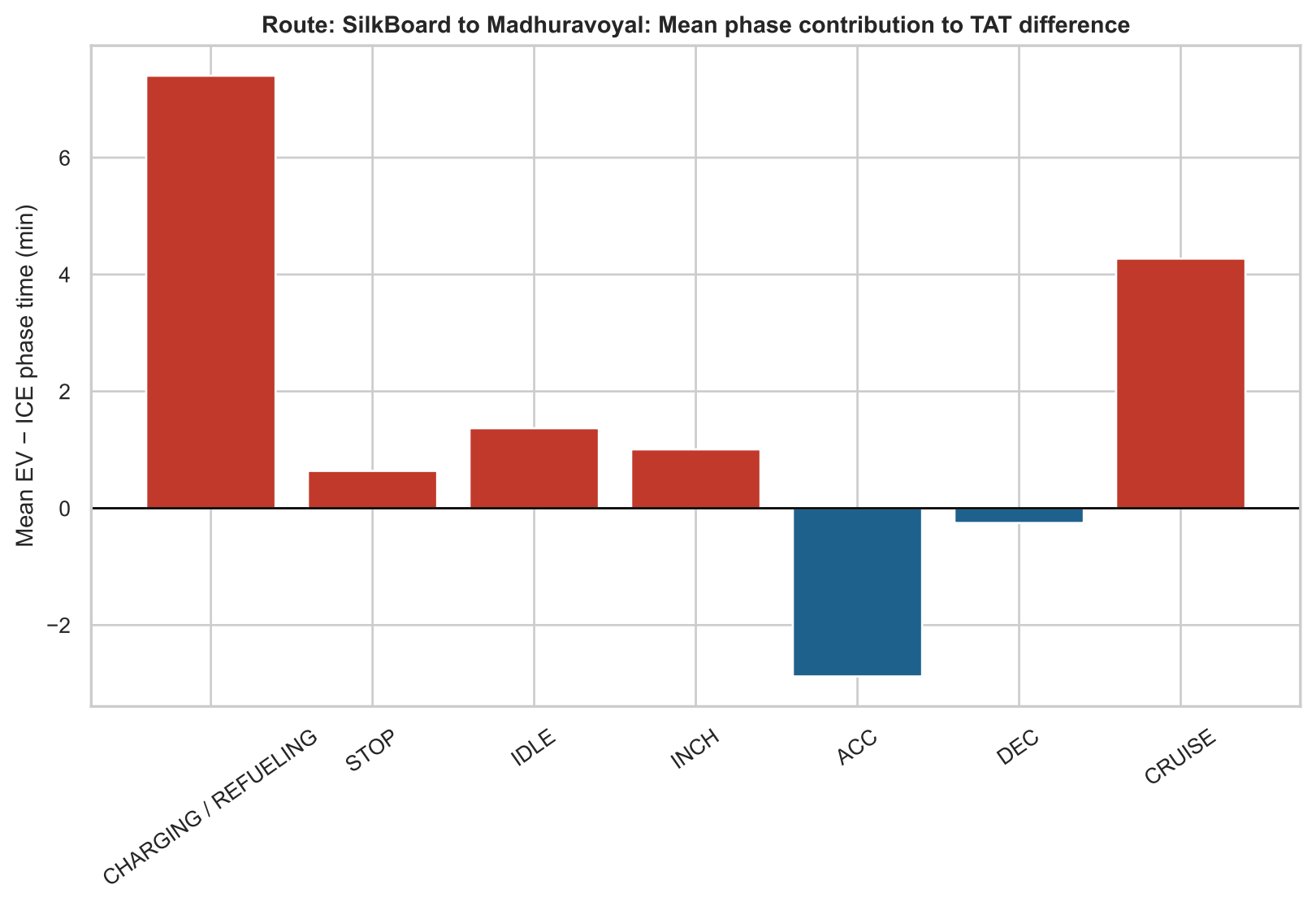
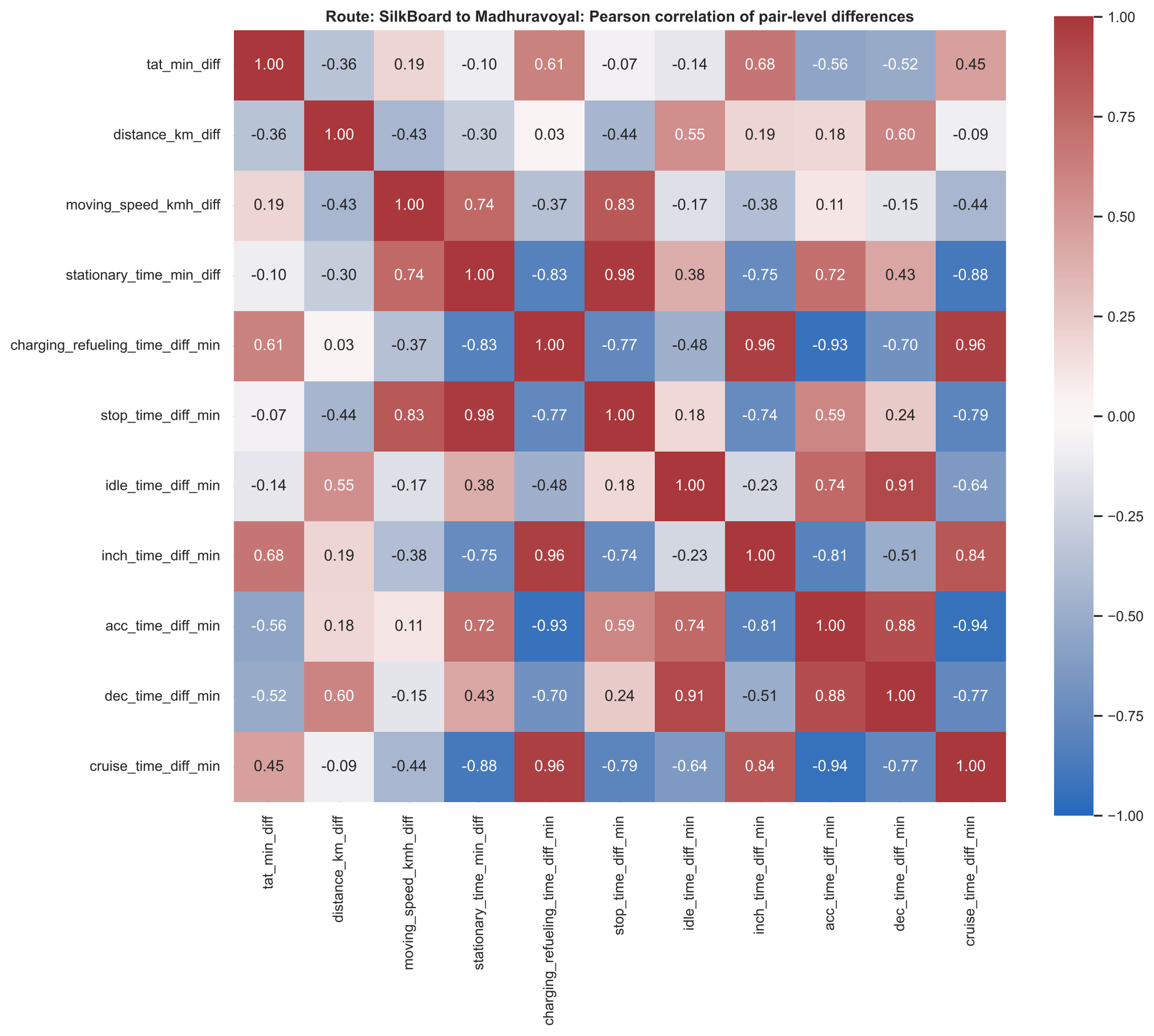

Route: SilkBoard to Madhuravoyal: Ranked pair-level TAT difference
PAIR_030
PAIR_032
PAIR_027
PAIR_001
PAIR_002
PAIR_028
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
EV TAT − ICE TAT (min)

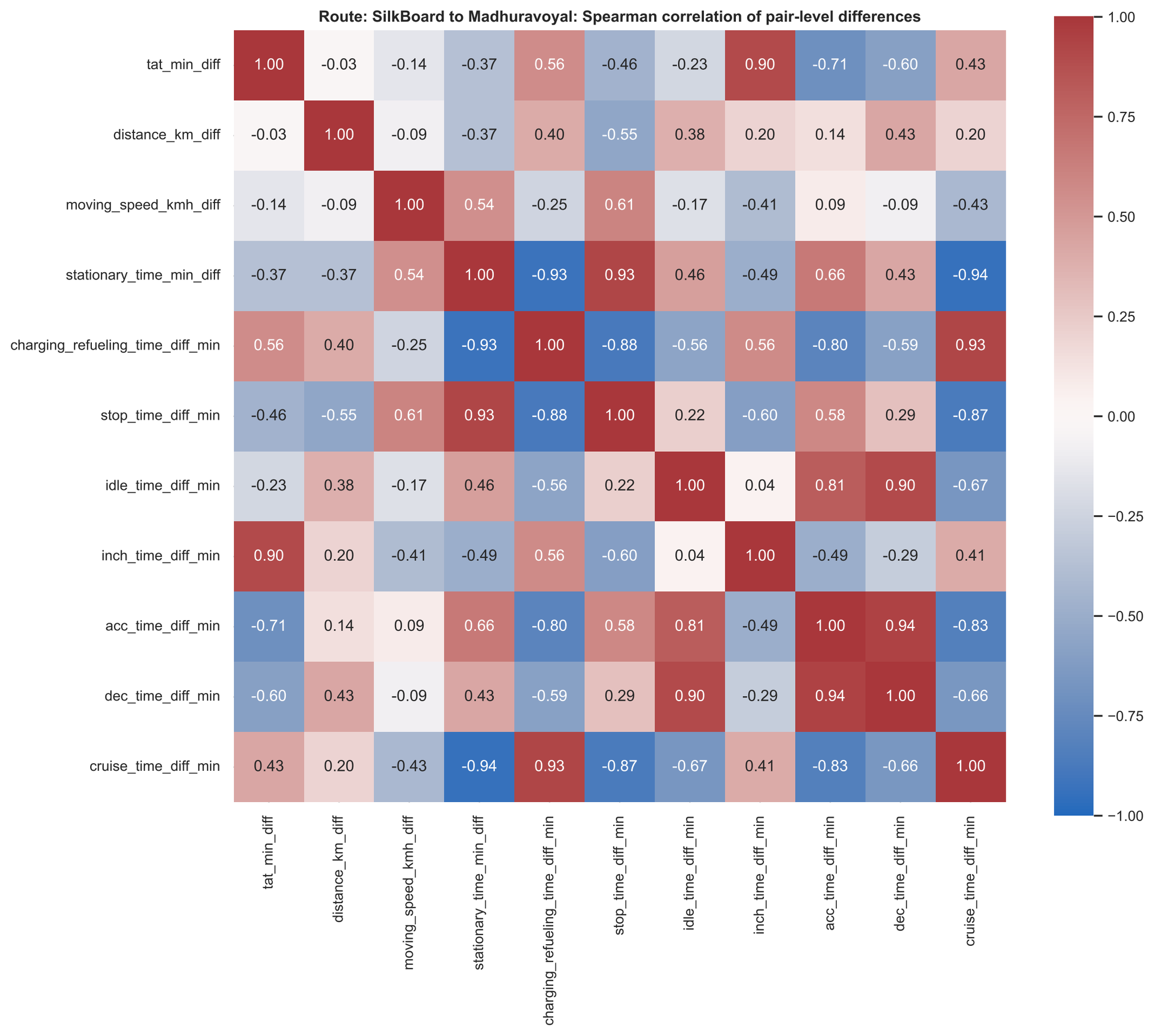

Route: SilkBoard to Madhuravoyal: Distribution of TAT differences
Count
3.0
2.5
2.0
1.5
1.0
0.5
0.0
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
EV TAT − ICE TAT (min)

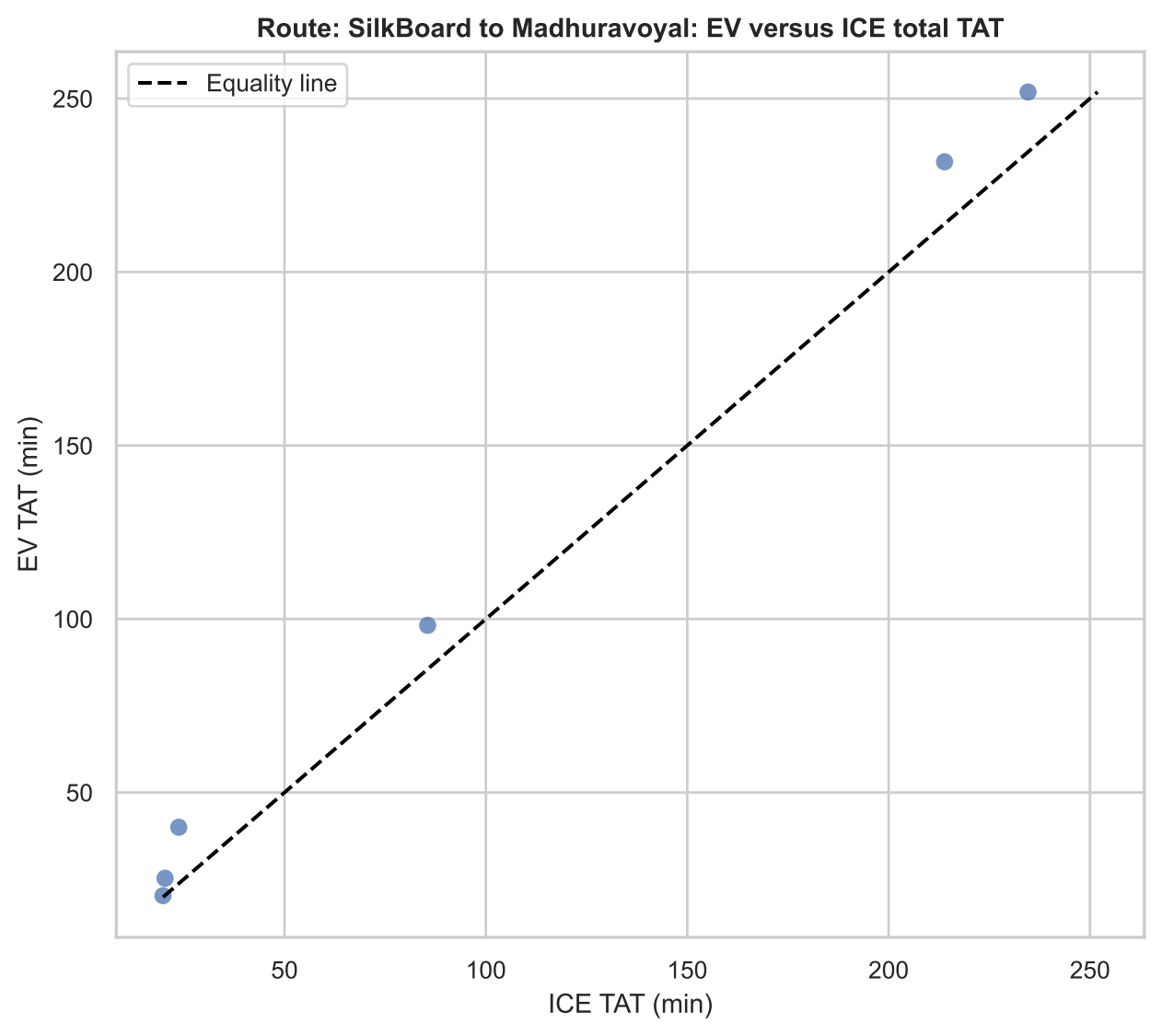

Route: SilkBoard to Madhuravoyal: EV and ICE TAT distribution
Total TAT (min)
250
200
150
100
50
EV
ICE

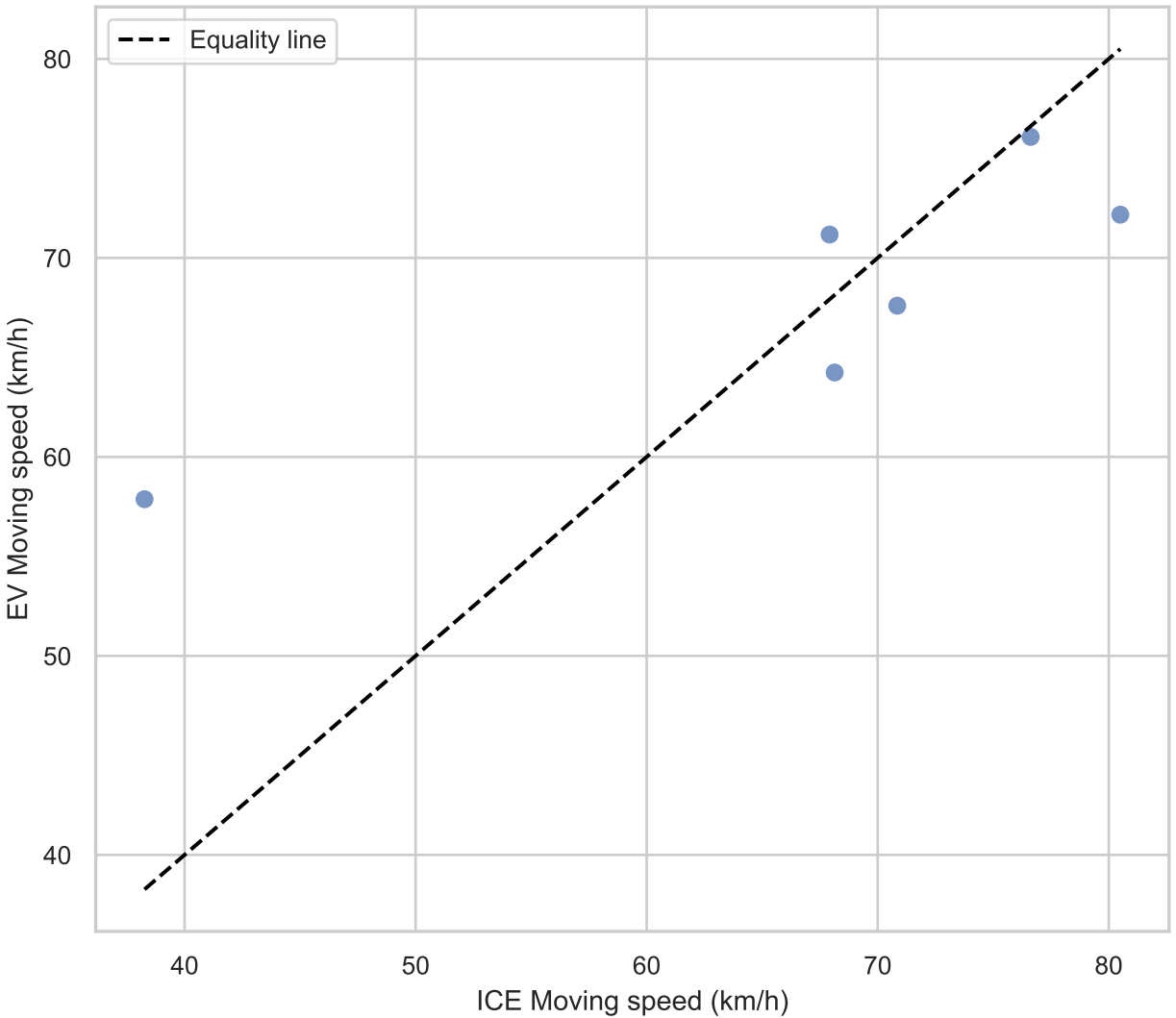

Route: SilkBoard to Madhuravoyal: Pair-by-phase time difference
pair_id
PAIR_028
PAIR_002
PAIR_001
PAIR_027
PAIR_032
PAIR_030
CHARGING / REFUELING
STOP
IDLE
INCH
ACC
DEC
CRUISE
EV − ICE phase time (min)
20
15
10
5
0
−5
−10
−15
−20

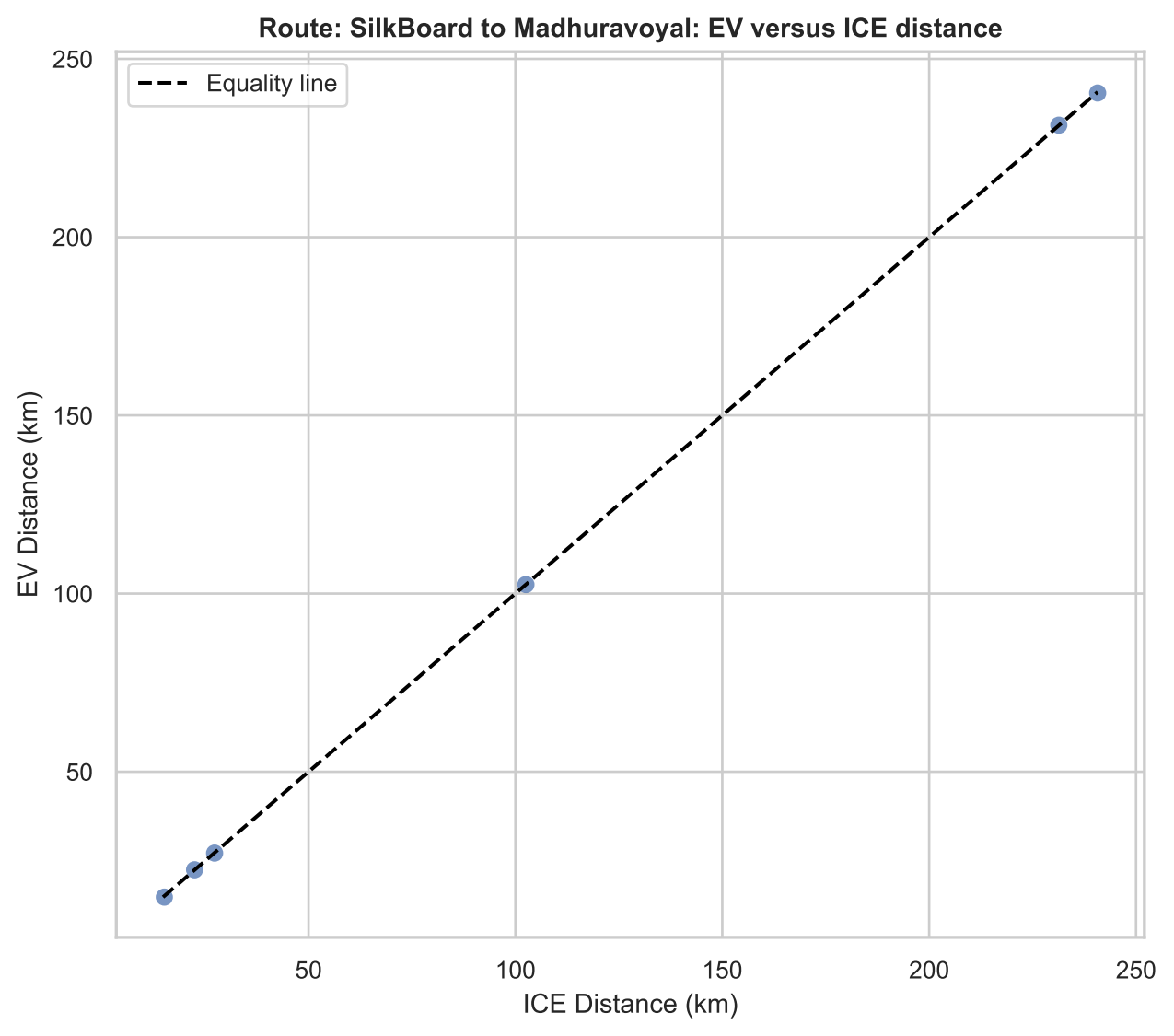

Route: SilkBoard to Madhuravoyal: Mean phase contribution to TAT difference
Mean EV – ICE phase time (min)
6
4
2
0
−2
CHARGING / REFUELING
STOP
IDLE
INCH
ACC
DEC
CRUISE

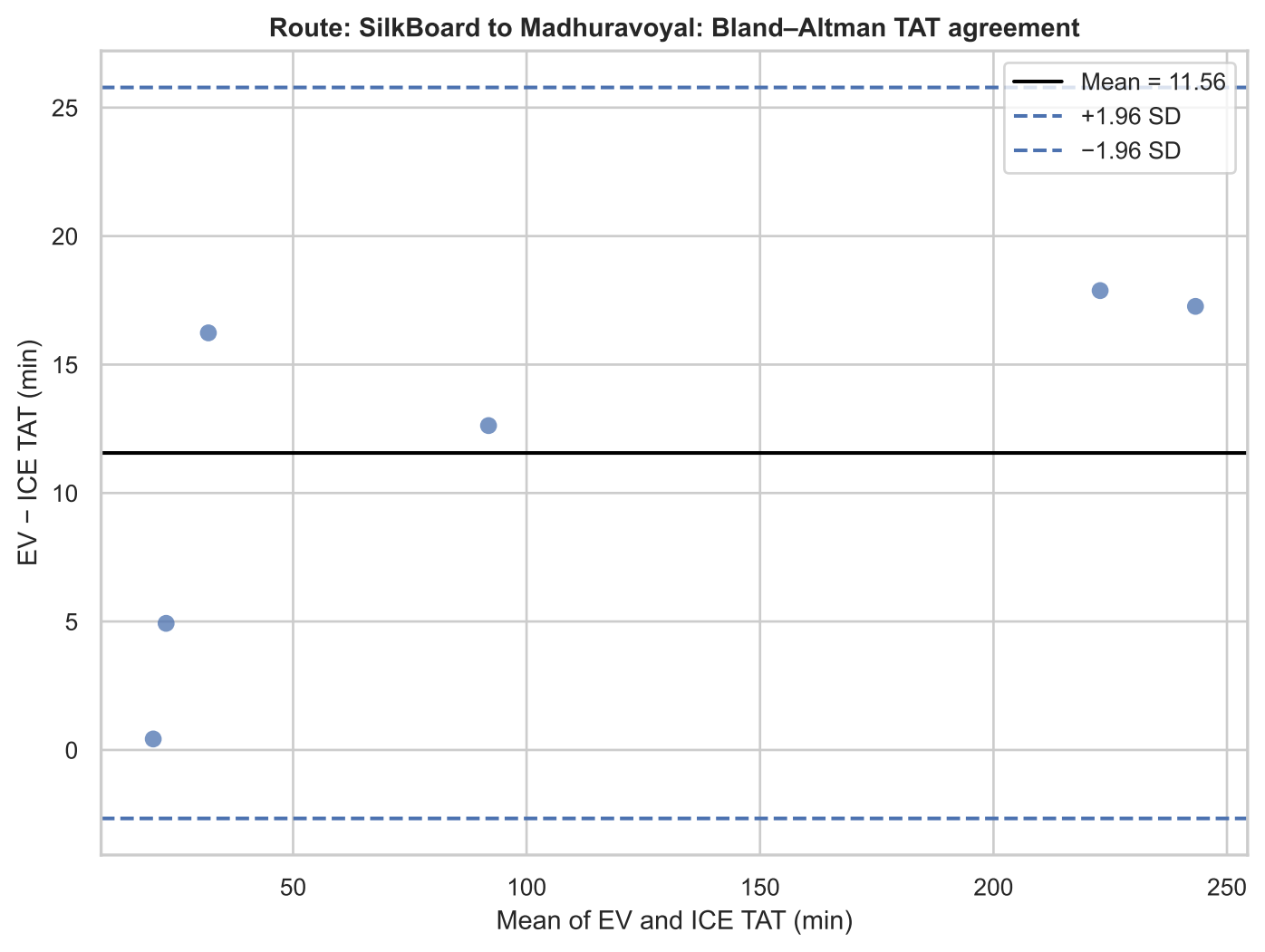

Route: SilkBoard to Madhuravoyal: Pearson correlation of pair-level differences
tat_min_diff 1.00 -0.36 0.19 -0.10 0.61 -0.07 -0.14 0.68 -0.56 -0.52 0.45
distance_km_diff -0.36 1.00 -0.43 -0.30 0.03 -0.44 0.55 0.19 0.18 0.60 -0.09
moving_speed_kmh_diff 0.19 -0.43 1.00 0.74 -0.37 0.83 -0.17 -0.38 0.11 -0.15 -0.44
stationary_time_min_diff -0.10 -0.30 0.74 1.00 -0.83 0.98 0.38 -0.75 0.72 0.43 -0.88
charging_refueling_time_diff_min 0.61 0.03 -0.37 -0.83 1.00 -0.77 -0.48 0.96 -0.93 -0.70 0.96
stop_time_diff_min -0.07 -0.44 0.83 0.98 -0.77 1.00 0.18 -0.74 0.59 0.24 -0.79
idle_time_diff_min -0.14 0.55 -0.17 0.38 -0.48 0.18 1.00 -0.23 0.74 0.91 -0.64
inch_time_diff_min 0.68 0.19 -0.38 -0.75 0.96 -0.74 -0.23 1.00 -0.81 -0.51 0.84
acc_time_diff_min -0.56 0.18 0.11 0.72 -0.93 0.59 0.74 -0.81 1.00 0.88 -0.94
dec_time_diff_min -0.52 0.60 -0.15 0.43 -0.70 0.24 0.91 -0.51 0.88 1.00 -0.77
cruise_time_diff_min 0.45 -0.09 -0.44 -0.88 0.96 -0.79 -0.64 0.84 -0.94 -0.77 1.00
tat_min_diff
distance_km_diff
moving_speed_kmh_diff
stationary_time_min_diff
charging_refueling_time_diff_min
stop_time_diff_min
idle_time_diff_min
inch_time_diff_min
acc_time_diff_min
dec_time_diff_min
cruise_time_diff_min
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00

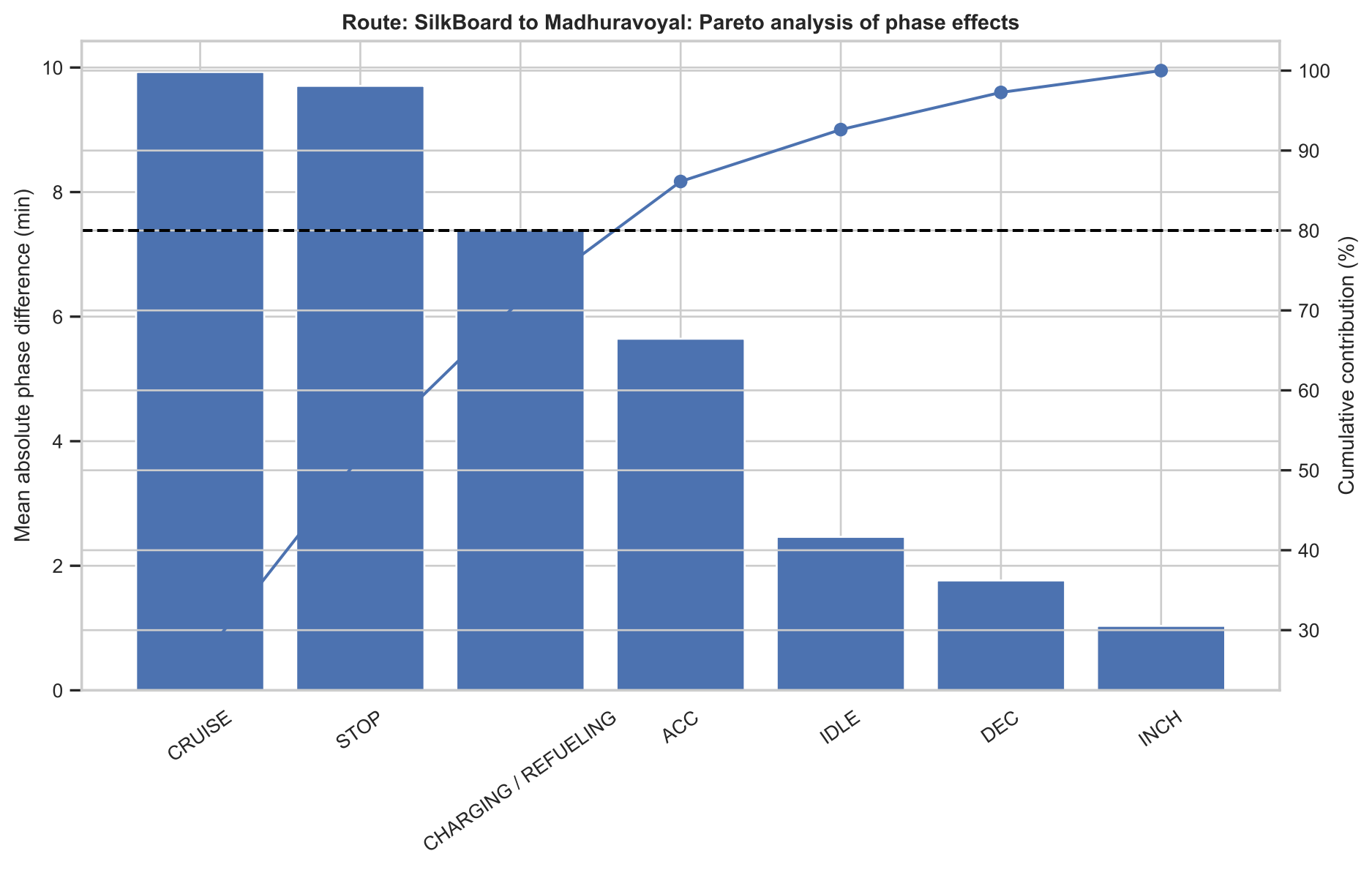

Route: SilkBoard to Madhuravoyal: Spearman correlation of pair-level differences
tat_min_diff 1.00 -0.03 -0.14 -0.37 0.56 -0.46 -0.23 0.90 -0.71 -0.60 0.43
distance_km_diff -0.03 1.00 -0.09 -0.37 0.40 -0.55 0.38 0.20 0.14 0.43 0.20
moving_speed_kmh_diff -0.14 -0.09 1.00 0.54 -0.25 0.61 -0.17 -0.41 0.09 -0.09 -0.43
stationary_time_min_diff -0.37 -0.37 0.54 1.00 -0.93 0.93 0.46 -0.49 0.66 0.43 -0.94
charging_refueling_time_diff_min 0.56 0.40 -0.25 -0.93 1.00 -0.88 -0.56 0.56 -0.80 -0.59 0.93
stop_time_diff_min -0.46 -0.55 0.61 0.93 -0.88 1.00 0.22 -0.60 0.58 0.29 -0.87
idle_time_diff_min -0.23 0.38 -0.17 0.46 -0.56 0.22 1.00 0.04 0.81 0.90 -0.67
inch_time_diff_min 0.90 0.20 -0.41 -0.49 0.56 -0.60 0.04 1.00 -0.49 -0.29 0.41
acc_time_diff_min -0.71 0.14 0.09 0.66 -0.80 0.58 0.81 -0.49 1.00 0.94 -0.83
dec_time_diff_min -0.60 0.43 -0.09 0.43 -0.59 0.29 0.90 -0.29 0.94 1.00 -0.66
cruise_time_diff_min 0.43 0.20 -0.43 -0.94 0.93 -0.87 -0.67 0.41 -0.83 -0.66 1.00
tat_min_diff
distance_km_diff
moving_speed_kmh_diff
stationary_time_min_diff
charging_refueling_time_diff_min
stop_time_diff_min
idle_time_diff_min
inch_time_diff_min
acc_time_diff_min
dec_time_diff_min
cruise_time_diff_min
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00

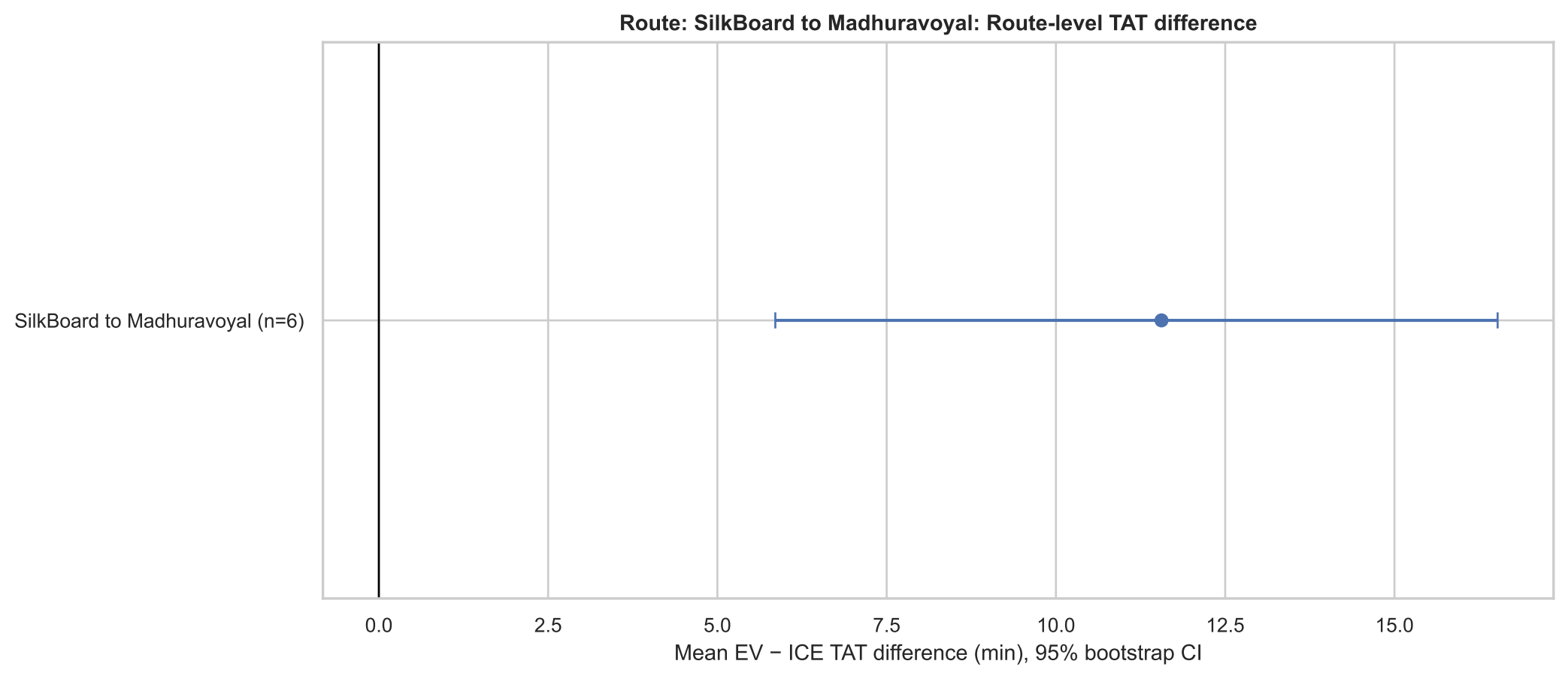

Route: SilkBoard to Madhuravoyal: EV versus ICE total TAT
Equality line
EV TAT (min)
ICE TAT (min)
50
100
150
200
250

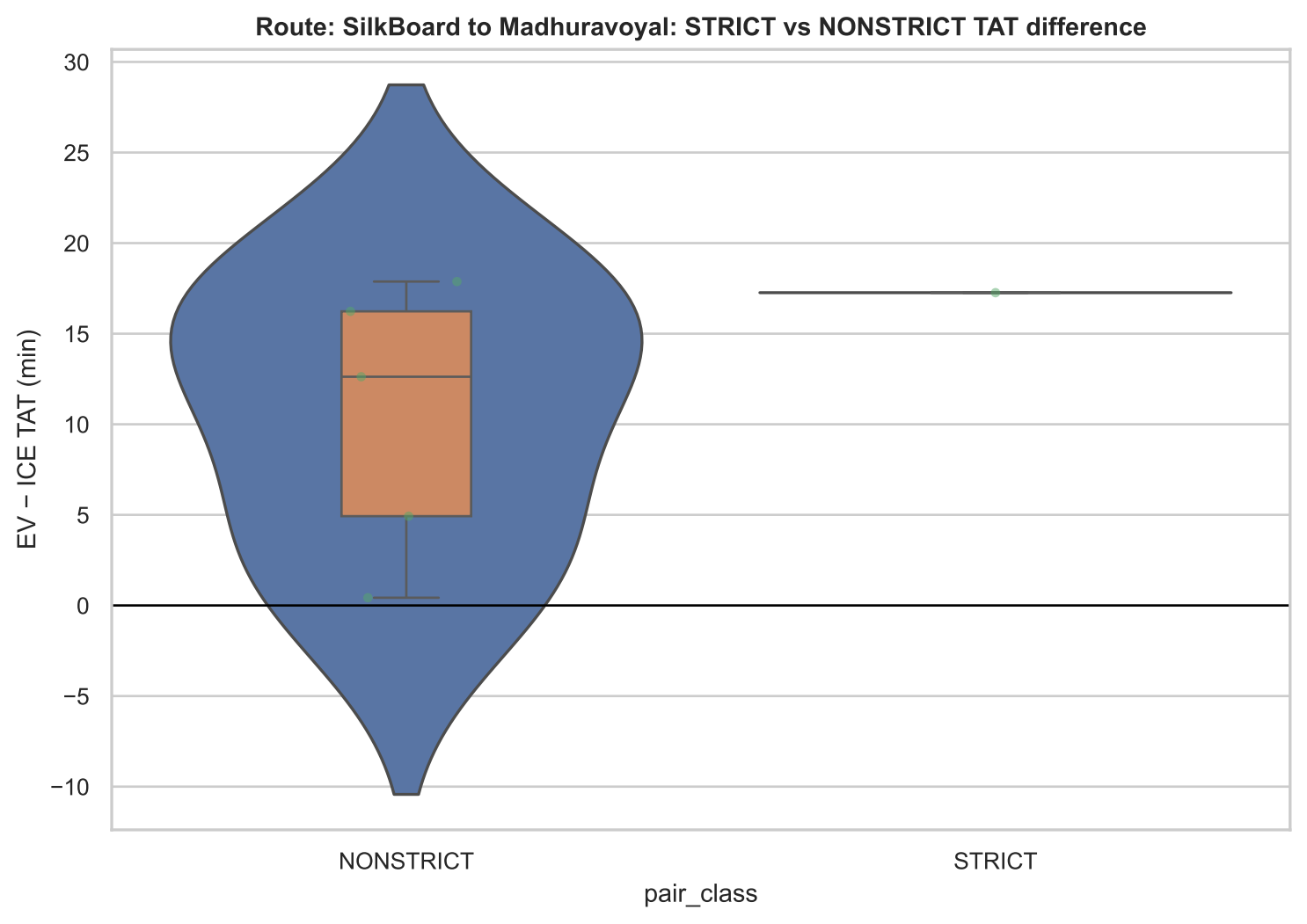

Route: SilkBoard to Madhuravoyal: EV versus ICE moving speed
Equality line
EV Moving speed (km/h)
ICE Moving speed (km/h)
40
50
60
70
80

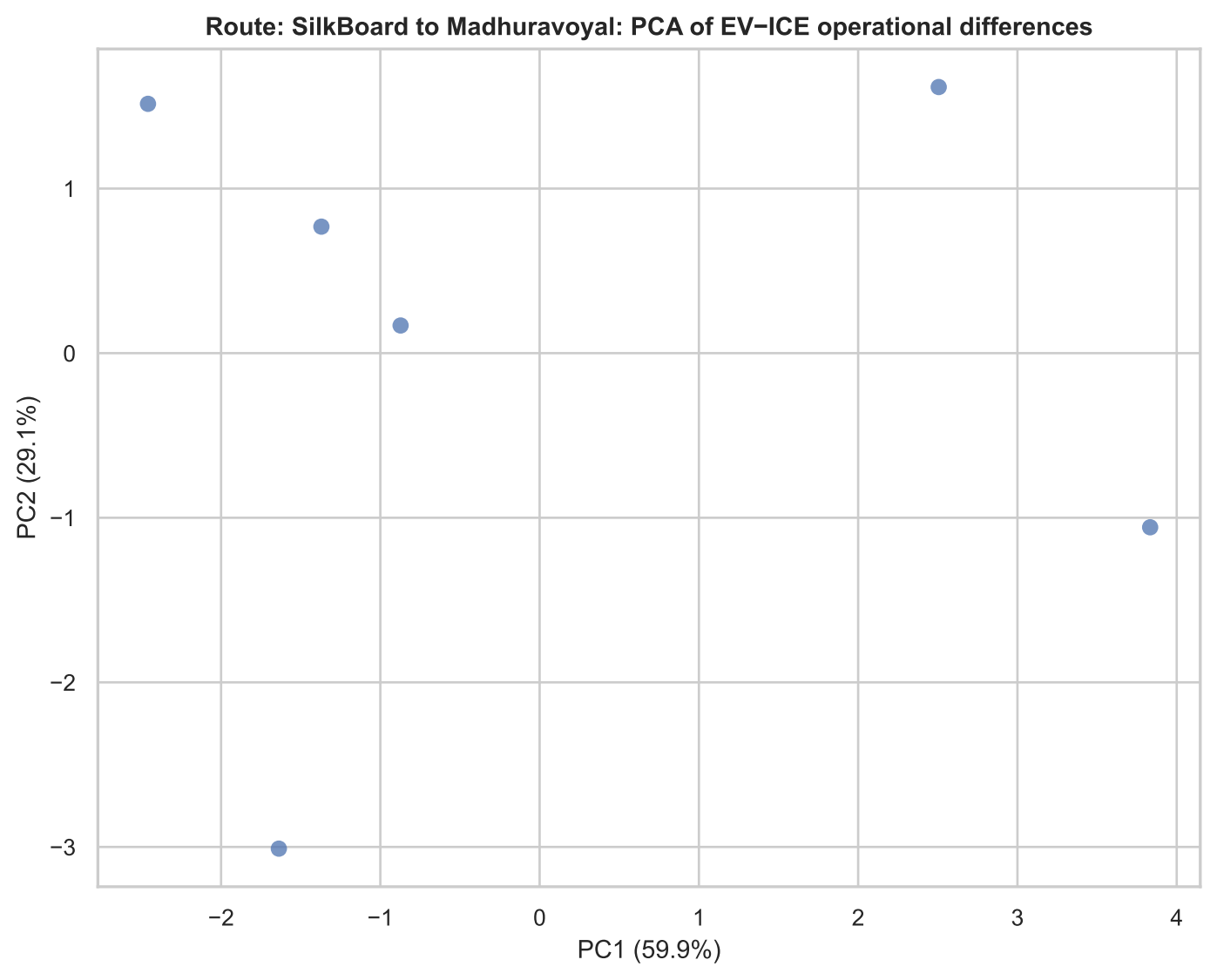

Route: SilkBoard to Madhuravoyal: EV versus ICE distance
Equality line
EV Distance (km)
ICE Distance (km)
250
200
150
100
50
50
100
150
200
250

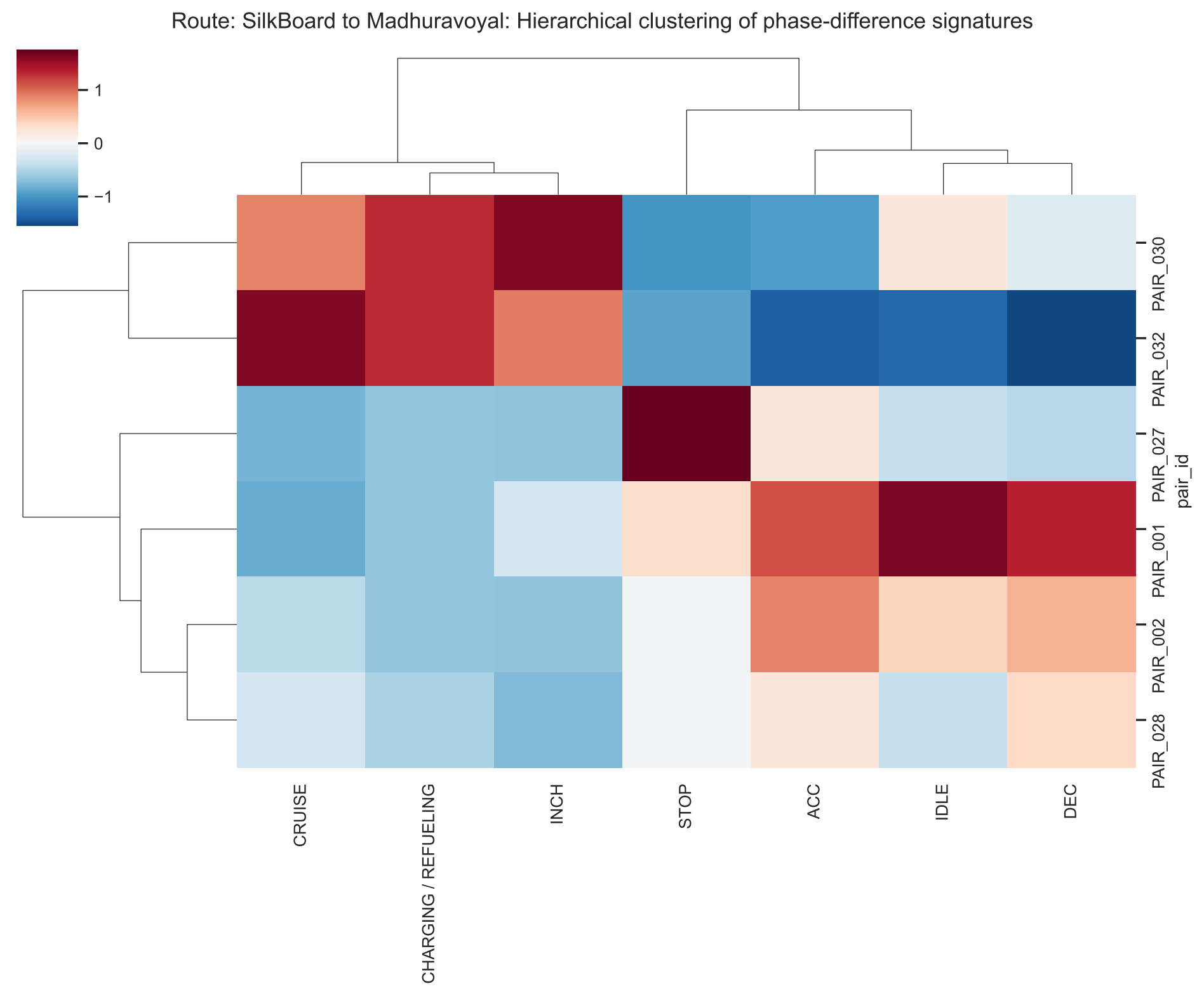

Route: SilkBoard to Madhuravoyal: Bland–Altman TAT agreement
Mean = 11.56
+1.96 SD
−1.96 SD
EV – ICE TAT (min)
Mean of EV and ICE TAT (min)
25
20
15
10
5
0
50
100
150
200
250


Route: SilkBoard to Madhuravoyal: Pareto analysis of phase effects
Mean absolute phase difference (min)
Cumulative contribution (%)
0
2
4
6
8
10
30
40
50
60
70
80
90
100
CRUISE
STOP
CHARGING / REFUELING
ACC
IDLE
DEC
INCH


Route: SilkBoard to Madhuravoyal: Route-level TAT difference
SilkBoard to Madhuravoyal (n=6)
0.0
2.5
5.0
7.5
10.0
12.5
15.0
Mean EV − ICE TAT difference (min), 95% bootstrap CI


Route: SilkBoard to Madhuravoyal: STRICT vs NONSTRICT TAT difference
EV – ICE TAT (min)
30
25
20
15
10
5
0
−5
−10
NONSTRICT
STRICT
pair_class


Route: SilkBoard to Madhuravoyal: PCA of EV−ICE operational differences
PC2 (29.1%)
PC1 (59.9%)
1
0
−1
−2
−3
−2
−1
0
1
2
3
4


Route: SilkBoard to Madhuravoyal: Hierarchical clustering of phase-difference signatures
1
0
−1
PAIR_030
PAIR_032
PAIR_027
PAIR_001
PAIR_002
PAIR_028
pair_id
CRUISE
CHARGING / REFUELING
INCH
STOP
ACC
IDLE
DEC